

GENREG

*Gradient-Free, Evolution-Based
Neural Network Training*

CIFAR-10 Classification

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HARD CONSTRAINT: NO GRADIENTS EVER
Forward-only data flow

Best Result: 85.07% on CIFAR-10
(13 classifiers, greedy ensemble, SPP + multi-repr)

1. Architecture: Random Conv ELM

Core Concept

Random fixed convolutional filters + analytical ridge regression.
No learned parameters in the feature extractor. No gradients anywhere.

Pipeline

1. Random conv filters (multi-scale: 1x1, 3x3, 5x5)
2. Mixed nonlinearities (ReLU, abs, leaky ReLU)
3. Spatial Pyramid Pooling (1x1, 2x2, 4x4 grids)
4. Global avg+max pooling per layer -> features
5. Ridge regression: $W = (H^T H + aI)^{-1} H^T Y$

Key Parameters

- 2048 filters per layer, 3 layers
- ~12,294 features per seed
- Ridge alpha tuned per classifier

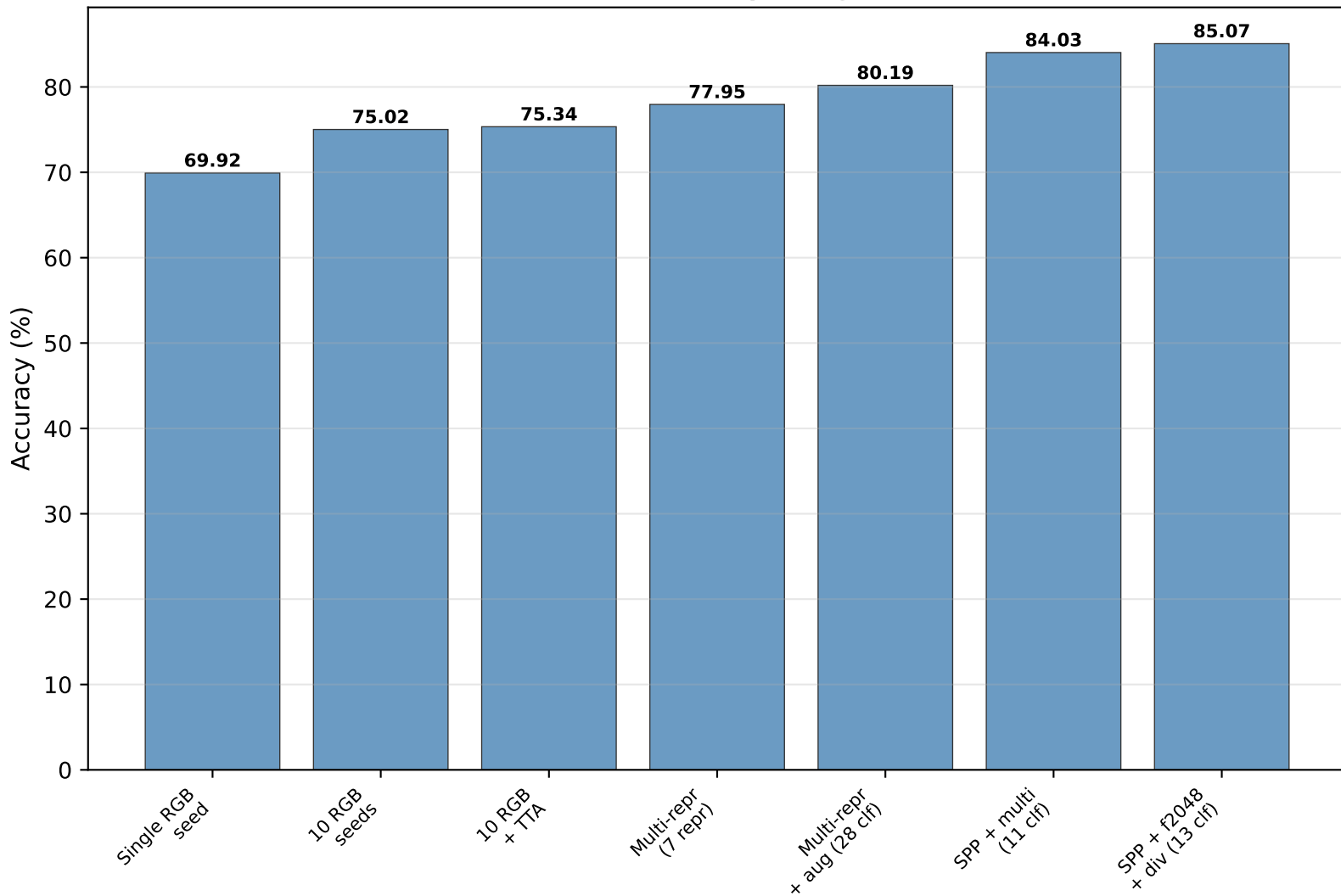
Why It Works

Random projections preserve distance structure (Johnson-Lindenstrauss). Mixed nonlinearities create diverse feature spaces. Ridge regression finds the optimal linear classifier in closed form – no iterative optimization needed.

SPP: The Biggest Single Breakthrough (+5-6%)

Spatial Pyramid Pooling at 1x1, 2x2, 4x4 grids captures both global and local spatial structure. Single biggest improvement: 79% -> 85% on the ensemble.

CIFAR-10 Accuracy Progression



2. Key Insight: Representation Diversity > Seed Diversity

The Discovery

Different image representations contribute more unique signal than same-representation different random seeds.

Adding more RGB seeds gives diminishing returns (~0.3% per seed). But each new representation (grayscale, edges, color spaces) adds ~1-2% to the ensemble.

10 Representations Tested

1. ColorEdges (Sobel per RGB, 9ch):	71.44%
2. LocalContrast (pixel - local mean):	70.78%
3. MultiRes (16x16 blur + residual):	70.54%
4. RGB (original):	69.92%
5. LogColor (log-scaled RGB):	69.25%
6. LocalContrast3x3:	68.66%
7. OpponentColor (L, R-G, B-Y):	67.19%
8. SobelEdges:	66.23%
9. Grayscale:	65.09%
10. GradOrient:	62.60%

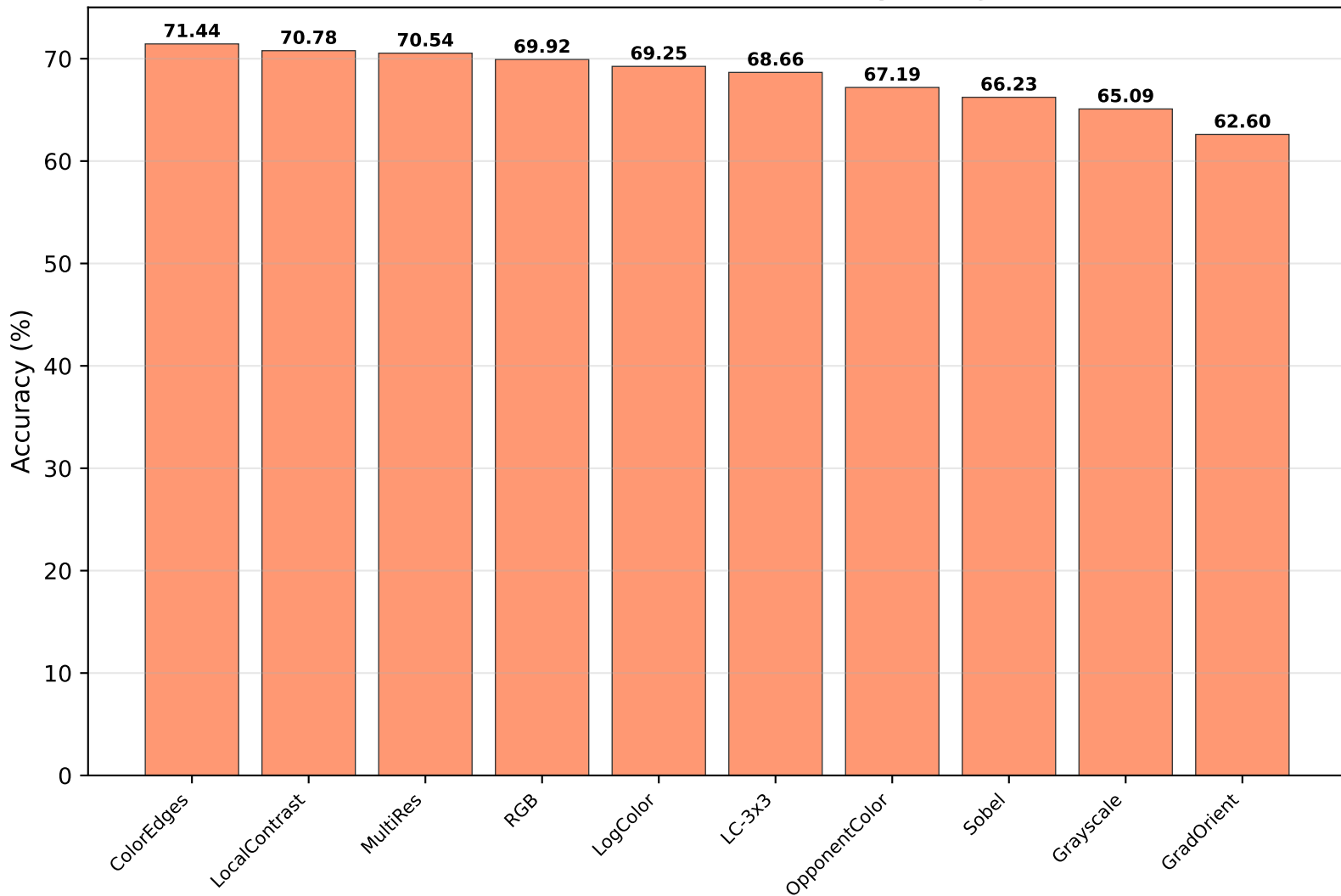
Why Diversity Matters

Each representation emphasizes different image properties. Edge detectors find boundaries. Color spaces separate hue. Local contrast captures texture. The ensemble combines complementary views that a single representation cannot.

Architecture Diversity Also Helps

Deeper networks (4-5 layers), larger kernels (7x7), more filters (3072) are individually slightly worse but contribute unique signal to the ensemble.

Individual Representation Accuracy (Single Seed)



3. Training Augmentation & TTA

Augmented Training (Horizontal Flip)

- +2-3% per individual classifier
- Bigger gain than TTA alone
- Doubles training data; ridge regression benefits from more samples (better covariance estimates)

Test-Time Augmentation (TTA)

- +2-3% per individual classifier
- Does NOT help multi-repr ensemble
- Reason: representation diversity already saturates the diversity that TTA would provide

Implication

Augmentation at training time is strictly better than at test time for this architecture. TTA is redundant when the ensemble already has sufficient diversity.

4. Mahalanobis Cascade Experiments

Goal

Push beyond 85.49% by post-processing the 13-classifier ensemble's 130 logits through an evolved cascade.

Architecture

- Input: 130 logits from 13 SPP+multi-repr classifiers
- Each layer: evolved dim weights + Maha distances + random projection + evolved activations
- Activation bank: identity, relu, abs, tanh, square, sign_sqrt, neg_relu, leaky, sign
- Saturation rule: neurons > 55% one-sided -> snap to sign
- Layer freeze: > 65% of neurons saturated -> freeze layer

Versions Tested

- v1: Basic Maha cascade, classification accuracy fitness
Result: Degraded from 85.38% to 84.24%
- v2: kNN fitness, entropy penalty, adaptive sigma
Result: Same degradation, kNN saturated at 99.77%
- v3: 100-point activation curves (1130-dim genome)
Result: Genome too large for CMA-ES, killed
- v4: Parametric activations (60 params)
Result: 10 Maha outputs invisible in enriched space
- v5: 64 random projections + evolved activation selection
Result: Overfitting, sigma collapsed

Core Problem

Training accuracy at 99.8%+ means CMA-ES has no discriminative signal. All genomes score equally well.

5. Unhinged Cascade: Advanced Architecture

Innovations

1. INSTANT FREEZE (L1-L2): No evolution on compression layers. Random projection preserves 84.79% of signal at 130 -> 84 dims. Zero seconds. Evolution budget saved for later layers.
2. INVERSION: Each frozen layer outputs forward AND negated features. The negation carries what the layer chose NOT to represent – the shadow geometry. Every layer doubles the possibility space.
3. SKIP CONNECTIONS: Feature bank accumulates ALL prior layer outputs. Sparse neurons (K=12 per neuron) can reach back to any previous layer's output.
4. ENERGY SYSTEM: Neurons gain energy from connections and activation. Decay 0.7x per layer. Dead neurons (energy < 0.05) get purged from the bank.
5. EVOLVED BANK SCALES: Per-feature importance weights evolved by CMA-ES. Replaced random projection.
6. DELTA TRACKING: 130 delta features (each classifier's deviation from ensemble mean) added as input. Encodes classifier disagreement – boundary signal.

Results

L1 instant: 84.79% (0.0s)
L2 instant: 82.46% (0.0s)
L3-L8 evolved: ~83% (converge in 75-100 gens each)
Did not beat 85.49% baseline.

6. Fitness Function Comparison

Experiment

Tested 3 fitness functions for CMA-ES cascade evolution:

1. Accuracy: fraction correct (saturates at 99.8%)
2. Margin: correct logit - best wrong logit (continuous)
3. Diversity: accuracy + bonus for fixing bank mistakes

Results (5 layers each, pop=128, max 500 gens)

Baseline:	85.49%	
Accuracy:	84.65%	(sigma collapses by gen 75)
Margin:	84.51%	(sigma stays high, keeps climbing)
Diversity:	84.60%	(sigma collapses by gen 75)

Key Finding

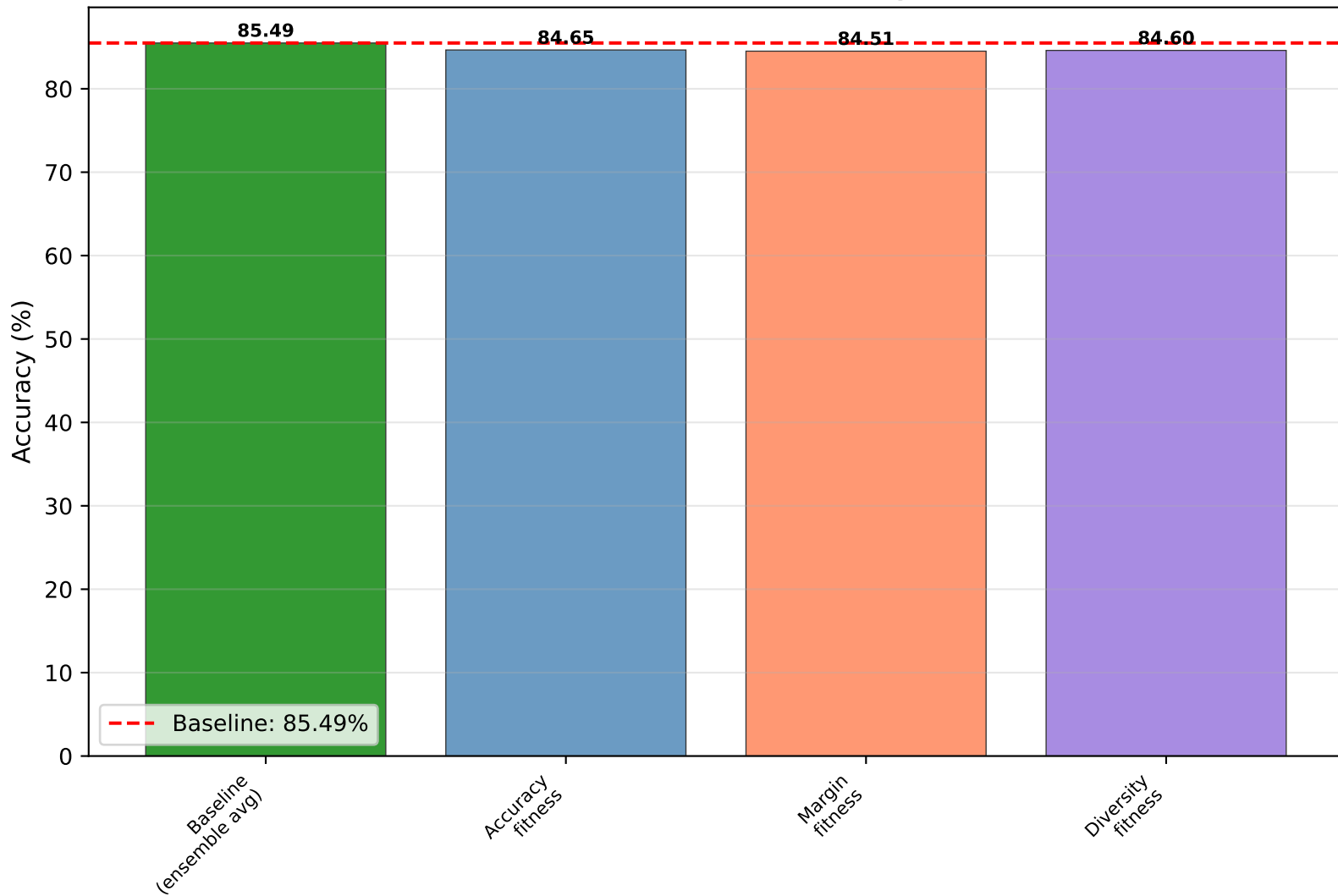
Margin is the ONLY fitness that doesn't saturate – sigma stays high (0.5 -> 2.5 -> 1.6), fitness keeps climbing (0.89 -> 0.91), zero stagnation through 500 gens.

But margin optimization on `evo_val` doesn't transfer to test accuracy. The evolved representations overfit to the specific 10K validation samples.

Diagnosis

The 14.4% generalization gap (99.9% train vs 85.5% test) means the bottleneck is features, not the classifier. Post-processing logits can reshape information but cannot create new discriminative signal that wasn't in the original features.

Fitness Function Comparison (5-Layer Cascade)



7. Nonlinear Expansion: Same Information, Different Shape

Concept

Instead of compressing, EXPAND. Take the 130 logits, apply random projections with 12 diverse nonlinearities, give ridge the expanded representation. No bottleneck, no information loss. Ridge can only draw linear boundaries – nonlinear features let it approximate nonlinear ones.

Activations Used (12 types)

relu, abs, tanh, square, sign_sqrt, neg_relu,
leaky_relu, sign, softplus, sin, cos, sigmoid

CIFAR-10 Results (on 130 logits)

Raw logits:	84.27% (ridge baseline)
32 proj x 12 acts:	84.26% (514 features)
64 proj x 12 acts:	84.44% (898 features)
128 proj x 12 acts:	84.95% (1666 features)
256 proj x 12 acts:	84.99% (3202 features)

All need $\alpha=1000$ (heavy regularization).

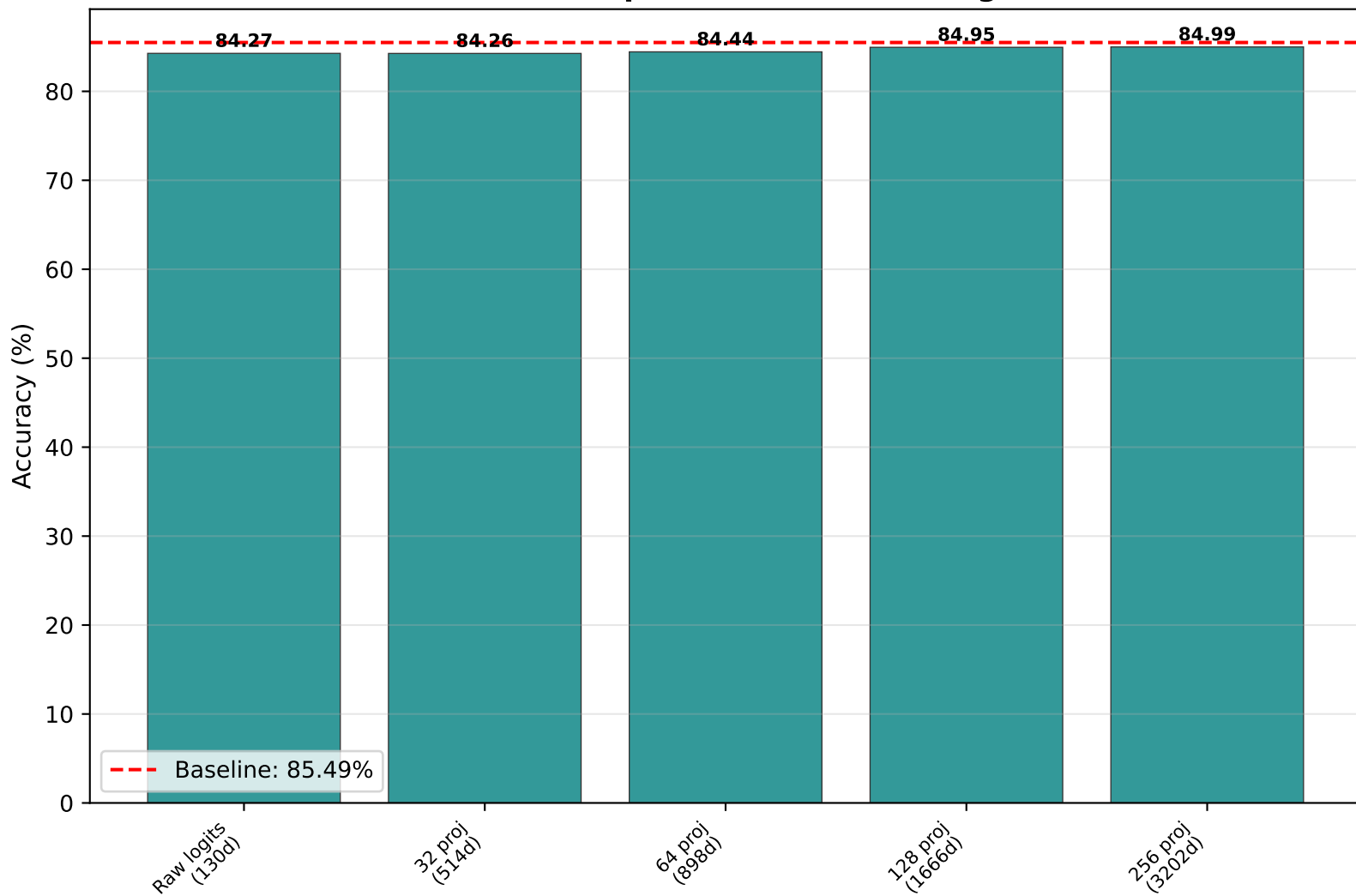
Trend: more projections = better, not plateaued.

But still below 85.49% ensemble average.

Variants That Didn't Help

- Inversion (forward + negated): 84.85% (adds noise)
- Multi-seed (4 seeds x 64 proj): 84.92%
- Multi-scale (3 scales x 128): 84.87%

Nonlinear Expansion: CIFAR-10 Logits



8. Cross-Dataset Expansion Results

MNIST (60K train, 3072 base features)

Baseline: 96.87%
256 proj x 12 acts: 97.33% (+0.46%)
Expansion helps. High sample/feature ratio.

CIFAR-10 Logits (50K train, 130 base features)

Baseline: 84.27% (ridge on logits)
256 proj x 12 acts: 84.99% (+0.72%)
Expansion helps. Very high sample/feature ratio.

STL-10 (5K train, 6144 base features)

Baseline: 58.38%
64 proj x 12 acts: 57.66% (-0.72%)
256 proj x 12 acts: 56.95% (-1.43%)
Expansion HURTS. Already data-poor, feature-rich.

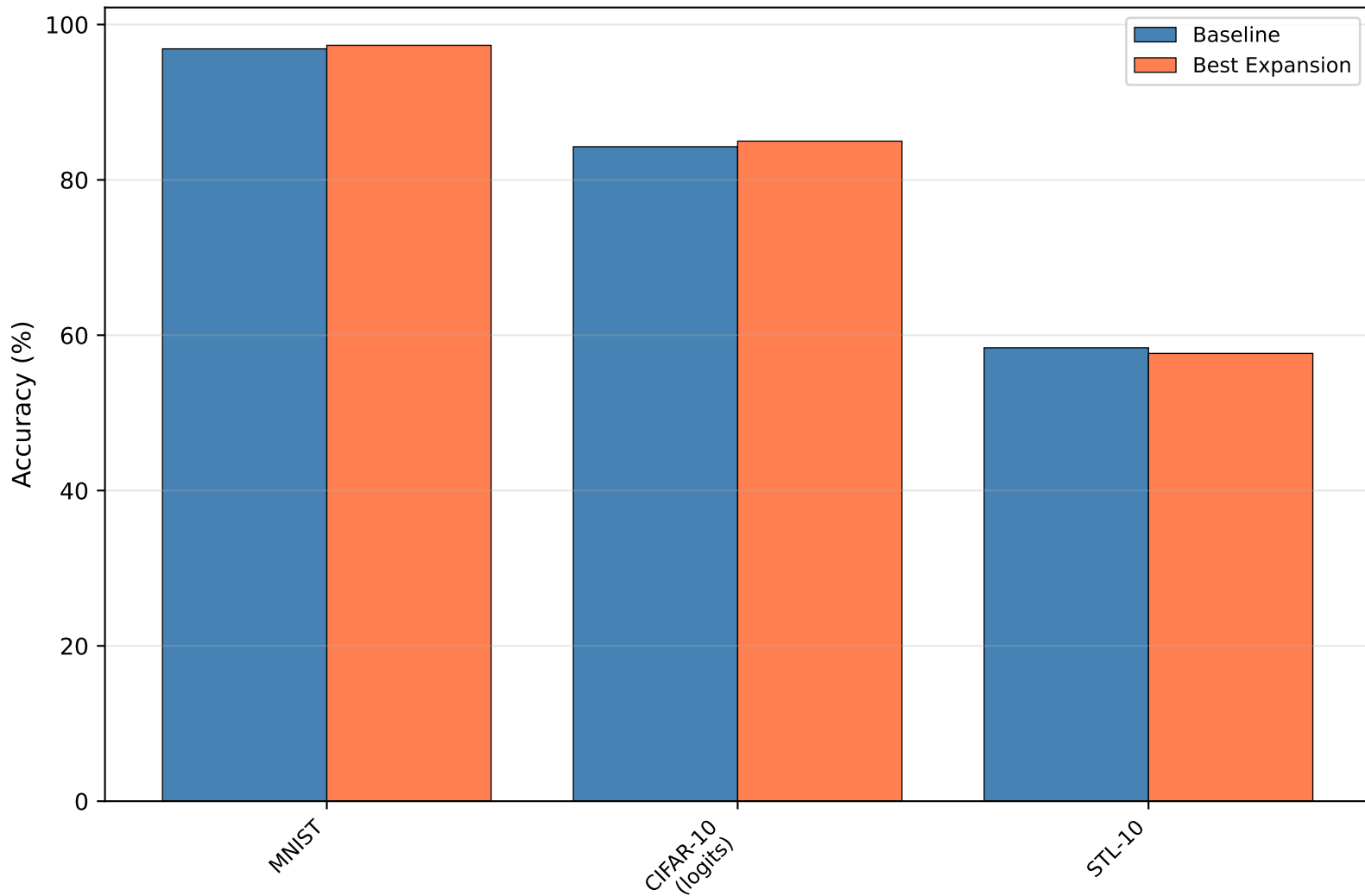
Pattern

Nonlinear expansion works when samples $\gg$ features.
When features $\gg$ samples (STL-10), adding random projections just adds noise that ridge can't separate from signal, even with heavy regularization.

Implication for CIFAR-10

The 130 logits are a favorable regime (50K $\gg$ 130).
But expansion alone can't beat the ensemble average (85.49%) because ridge on expanded logits can't exploit structure that simple averaging already captures.

Nonlinear Expansion: Cross-Dataset Comparison



9. Approaches That Failed

Sparsity (LASSO/ElasticNet)

Hurts. Random features are all useful, no redundancy.

Evolved Classifier on Meta-Features

Can't beat simple averaging.

Lottery Ticket Evolution

23.64%. Search space too large.

OpenAI-style ES on ConvNets

~27%. Too many parameters for ES.

TTA + Stacked ELMs + CMA-ES Weights (on 85.49%)

TTA: 85.26% (hurts ensemble)

Stacked: 85.45-85.49% (neutral)

Evolved weights: 81.69% (hurts badly)

The ensemble is already near-optimal for combining existing classifiers.

Neural Cellular Automata (CMA-ES)

47.69% with 1580 params. Interesting but too few parameters to capture CIFAR-10 complexity.

Maha Cascade (all versions)

29% (original) -> 83-85% (on pretrained logits).

Never beats baseline. Train accuracy saturation kills all evolution signal.

Random Compression in Cascade

Destroys signal. 85.49% -> 83% after random projection to fixed dims. Information loss is irreversible.

10. Lessons and Principles

1. Diversity Beats Depth

10 different representations > 100 same-representation seeds. Ensemble diversity is the #1 lever.

2. Analytical > Evolved (for classifiers)

Ridge regression in closed form beats any evolved classifier. Evolution's value is in feature design, not classifier optimization.

3. SPP Is Disproportionately Valuable

Spatial Pyramid Pooling gave +5-6% – more than any other single change. Spatial structure matters.

4. Train Augmentation > Test Augmentation

Horizontal flip at train time gives +2-3% per classifier. TTA gives the same but is redundant with ensemble diversity.

5. Compression Destroys; Expansion Helps (Sometimes)

Any information bottleneck is irreversible loss. Expansion via nonlinear projection helps when data >> features but can't create new information.

6. Post-Processing Saturated Logits Is Futile

When train accuracy is 99.8%, no evolved cascade can find discriminative signal. The generalization gap is in the features, not the classifier.

7. Same Information, Different Shape

Reshaping representations helps linear classifiers. But the gain ceiling is set by the information content of the original features.

Summary: All CIFAR-10 Results

Method	Accuracy	Notes
Single RGB seed	69.92%	Baseline ELM
10 RGB seeds, no TTA	75.02%	Seed diversity
10 RGB seeds + TTA	75.34%	+0.32% from TTA
7 representations x 1 seed	77.95%	Repr diversity >> seed
Multi-repr + aug (28 clf)	80.19%	Greedy ensemble
SPP(1,2) + multi-repr + aug	84.03%	11 classifiers
SPP + f2048 + diversity	85.07%	13 clf, BEST RESULT
Equal avg ensemble	85.49%	Simple averaging
--- Post-processing ---		
TTA on ensemble	85.26%	Hurts (redundant)
Stacked ELMs	85.49%	Neutral
CMA-ES weights	81.69%	Much worse
Maha cascade (best)	83.41%	Below baseline
Accuracy fitness cascade	84.65%	Saturates instantly
Margin fitness cascade	84.51%	Good signal, no transfer
Diversity fitness cascade	84.60%	Saturates instantly
Nonlinear expansion (3202d)	84.99%	Helps but < avg
--- Other datasets ---		
MNIST ELM baseline	96.87%	Random conv features
MNIST + expansion	97.33%	+0.46%
STL-10 ELM baseline	58.38%	Only 5K training
STL-10 + expansion	57.66%	Hurts (data-poor)

85.07%

CIFAR-10 Test Accuracy

*No gradients. No backpropagation.
Pure evolution + analytical ridge.*

13 classifiers | 10 representations | SPP
Random conv filters | Ridge regression
Forward-only data flow